

CHAPTER: 5

BACK END OF SYSTEM

5.1 About Backend

5.2 Backend Technologies

5.3 Core Backend Features

5.3.1 about MySQL

5.3.2 How MySQL Works

5.3.3 MySQL Features

5.4 Data Handling

5.5 Important Controller Groups

5.1 About Backend

CodeVerse backend follows layered architecture with Controllers, Services, Repositories, Entities, DTOs, Filters, and Utility classes.

5.2 Backend Technologies

- Java 17
- Spring Boot 4.0.2
- Spring MVC
- Spring Data JPA (Hibernate)
- MySQL Database
- Maven Build

5.3 Core Backend Features

- Session-based authentication with role checks
- BCrypt password hashing
- CSRF protection filter
- Role-based route guarding via AuthFilter
- Mail service for welcome and reset flows
- Payment processing integration

5.4 Database Handling

- Datasource URL configured for MySQL database codeversse
- JPA ddl-auto set to update
- SQL logging enabled for debugging

5.5 Important Controller Groups

- SessionController: authentication and account recovery
- AdminController: dashboard and administration
- Participant controllers: home, teams, applications
- Organizer controllers: applications, judge assignment, profile
- JudgeController: assignments and scorecards
- PaymentController: credit card charge flow